

# YOLOv8 Building Detection from Satellite

## Problem

This project builds an automated building detection system for satellite imagery using deep learning. Given aerial photos of Paris from the SpaceNet dataset, a YOLOv8 model is trained to locate and draw bounding boxes around individual buildings. This system is useful for urban planning, disaster response, population estimation, and infrastructure mapping — tasks where manually analyzing thousands of satellite images is time-consuming and impractical.

## Data

The dataset is from the SpaceNet Paris challenge (AOI 3), accessed via Kaggle. It contains **1,148 RGB satellite image tiles** (650×650 pixels each), cropped from high-resolution PanSharpen imagery over Paris, France. Each image is paired with a GeoJSON label file containing building footprint polygons in WGS84 geographic coordinates (longitude/latitude). Labels were converted to YOLO bounding box format (.txt) using rasterio to map geographic coordinates to pixel space. The dataset was split **80/10/10** into train (918 images), validation (115 images), and test (115 images) sets using sklearn's train\_test\_split.

## Model

We used **YOLOv8n (nano)**, the smallest and fastest variant of Ultralytics YOLOv8 — a single-stage real-time object detector that predicts bounding boxes and class probabilities in one forward pass. Key model specs:

| PARAMETER          | VALUE                     |
|--------------------|---------------------------|
| Architecture       | YOLOv8 nano (yolov8n)     |
| Total Parameters   | 3,005,843                 |
| Layers             | 73                        |
| Pretrained Weights | COCO dataset (yolov8n.pt) |
| Number of Classes  | 1 (building)              |

Transfer learning from COCO pretrained weights was used — the model already knows how to detect edges, shapes, and textures, so fine-tuning requires far fewer epochs than training from scratch

## Training

| HYPERPARAMETER | VALUE     |
|----------------|-----------|
| epochs         | 25        |
| Batch size     | 16        |
| Image size     | 640 * 640 |
| GPU            | Tesla T4  |
| Training Time  | 8 minutes |

## Observations

### What worked well:

Transfer learning from COCO weights gave a strong starting point. The model learned to detect buildings of varying sizes and orientations within just 25 epochs, achieving 73% mAP@50 with well-balanced precision and recall (F1 = 0.732).

### What the model struggled with:

Dense clusters of tightly-packed Parisian buildings caused over- or under-counting. Many tiles contained water, parks, or black padding (52/115 test images had zero buildings), creating class imbalance. Shadows, dark rooftops, and partially visible edge buildings also made accurate boundary detection harder.

### One idea to improve it:

Training for more epochs (50-100) with a larger model variant (YOLOv8s or YOLOv8m) would likely push mAP@50 above 0.80. Additionally, filtering out empty background tiles during training and adding instance segmentation (YOLOv8-seg) to capture precise building footprints rather than bounding boxes could significantly improve results.